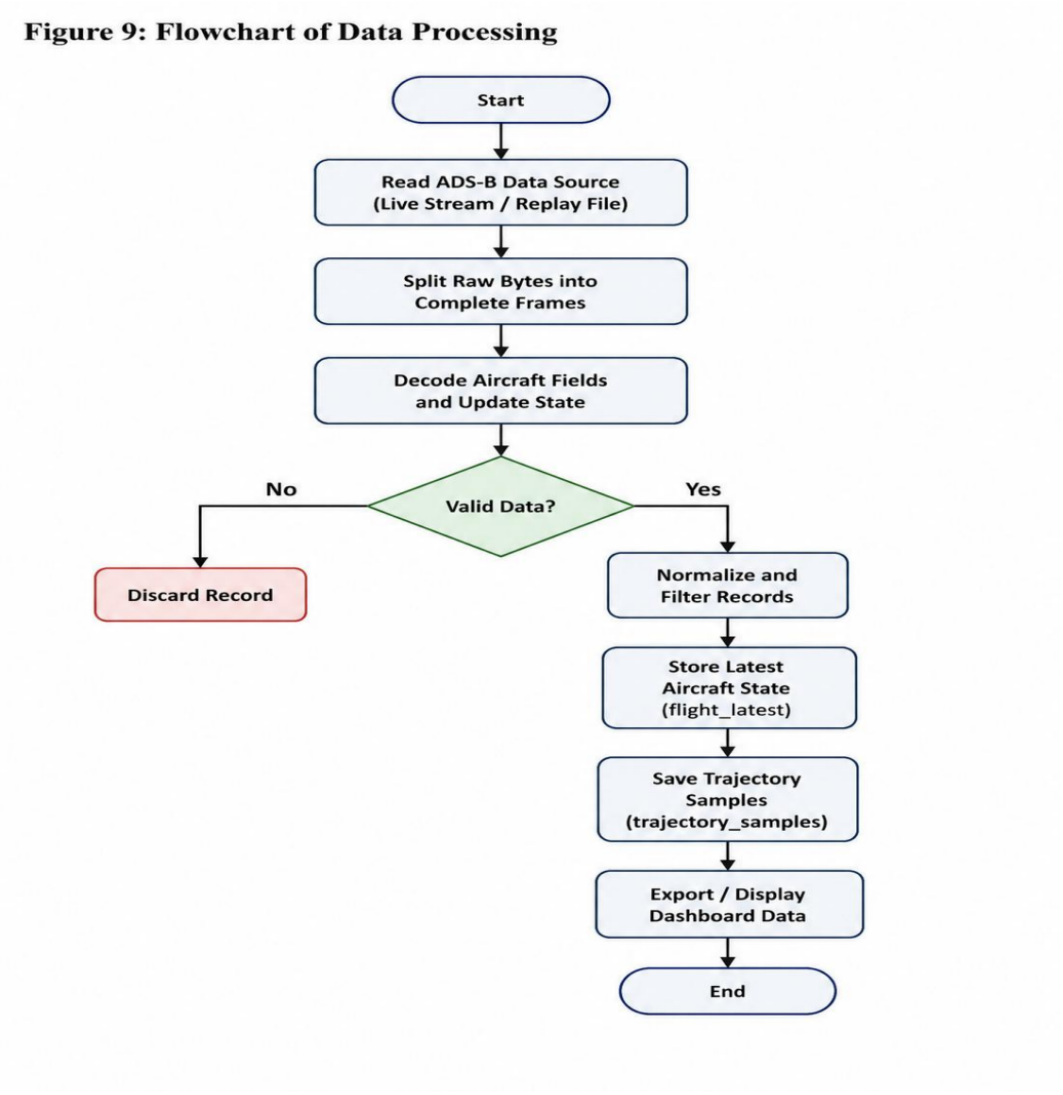



*Figure 9: Flowchart of Data Processing*

### 4.6 Design Rationale

A local Flask application was selected because the project needs simple API endpoints, not a large enterprise framework. SQLite was selected because it stores structured data in one file and is suitable for local academic projects. CSV and

**4.6 Design Rationale**

A local Flask application was selected because the project requires compact JSON APIs rather than a large enterprise framework [4]. SQLite provides embedded file-based storage [5], and WAL mode supports the project's local read/write pattern [6]. CSV and Excel-compatible exports make captured data easy to inspect with common analytical tools. Leaflet provides interactive map layers and markers [7], while Chart.js renders the selected aircraft's altitude history [8].